

SHIVANG AGGARWAL

shivang.aggarwal@hpe.com

<https://shivang94.github.io>

SUMMARY: Currently working as a Senior Research Scientist in the Networking and Distributed Systems Lab (NDSL) at HPE Labs with research interests in mobile computing and wireless networks such as Wi-Fi 6/7 (WLANs) and private/public 5G/6G (cellular)

EDUCATION

Northeastern University, Massachusetts, USA <i>Ph.D., Computer Engineering</i>	December 2022
University at Buffalo, State University of New York, New York, USA <i>Master of Science, Computer Science and Engineering</i>	December 2017
Manipal Institute of Technology, Karnataka, India <i>Bachelor of Technology, Computer Science and Engineering</i>	July 2016

EXPERIENCE

Senior Research Scientist – Networking & Distributed Systems Lab, HPE Labs • Working in the Networking and Distributed Systems Lab (NDSL) at Hewlett Packard Enterprise (HPE) Labs	Sept 2022 – Present
Research Assistant – WiNS Lab (Northeastern University/University at Buffalo) • Worked towards building and evaluating reliable, high-throughput mmWave (802.11ad) Wi-Fi systems for mobile devices	Oct 2017 – Aug 2022
Research Intern – NEC Laboratories America, Inc. • Worked on enabling real-time user localization and tracking in indoor environments leveraging existing Wi-Fi infrastructure using 802.11mc Wi-Fi FTM ranging	May 2021 – Dec 2021
Research Assistant – ODIn Lab (University at Buffalo) • Worked on the Mimir database, a probabilistic database that intelligently guesses missing and fixes incorrect values to efficiently manage messy data	Jun 2017 – Dec 2017
Research and Development Intern – Defence Research and Development Organisation (DRDO) • Used data mining techniques to find ways to improve the cyber security of government networks by increasing the rate of intrusion detections.	Feb 2016 – Jun 2016

SELECTED PUBLICATIONS

- **locRoaming: Seamless Private 5G WiFi Convergence through Continuous Client Positioning**
Yu-Tai Lin, Mohamed Ibrahim, **Shivang Aggarwal**, Haofan Lu, Puneet Sharma, Karthikeyan Sundaresan
ACM EdgeSys 2026
- **Beyond Wi-Fi: Private 5G for Better Multi-User Quality-of-Experience**
Ushasi Ghosh, **Shivang Aggarwal**, Mohamed Ibrahim, Puneet Sharma, Dinesh Bharadia
ACM HotMobile 2026
- **How Mature is 5G Deployment? A Cross-Sectional, Year-Long Study of 5G Uplink Performance**
Imran Khan, Moinak Ghoshal, Joana Angjo, Sigrid Dimce, Mushahid Hussain, Paniz Parastar, Yenchia Yu, Xueting Deng, Sumit Hawal, Shirui Huang, Ameya Rane, Yin Wang, Claudio Fiandrino, Charalambos Orfanidis, **Shivang Aggarwal**, Ana Aguiar, Ozgu Alay, Carla Fabiana Chiasserini, Falko Dressler, Y. Charlie Hu, Steven Ko, Dimitrios Koutsonikolas, and Joerg Widmer
IFIP/IEEE NETWORKING 2024 & Elsevier COMCOM 2025
- **Maestro: QoE-Aware Dynamic Resource Allocation in Wi-Fi Networks**
Umakant Kulkarni, Khaled Diab, Lianjie Cao, Faraz Ahmed, **Shivang Aggarwal**, Puneet Sharma, and Sonia Fahmy
ACM CoNEXT 2025
- **RFBridge: Ultra Wideband Reconfigurable Metamaterial Surface Enabling Frequency Conversion**
Yawen Liu, **Shivang Aggarwal**, Mohamed Ibrahim, Puneet Sharma, Swarun Kumar
ACM HotMobile 2025

- **Wixor: Dynamic TDD Policy Adaptation for 5G/xG Networks**
Ahmad Hassan, **Shivang Aggarwal**, Mohamed Ibrahim Ahmed, Puneet Sharma, Feng Qian
ACM CoNEXT 2024
- **Can 5G mmWave Enable Edge-Assisted Real-Time Object Detection for Augmented Reality?**
Moinak Ghoshal, Z. Jonny Kong, Qiang Xu, Zixiao Lu, **Shivang Aggarwal**, Imran Khan, Jiayi Meng, Yuanjie Li, Y. Charlie Hu, Dimitrios Koutsonikolas
IEEE MASCOTS 2023
- **Understanding the Impact of Wi-Fi Configuration on Volumetric Video Streaming Applications**
Umakant Kulkarni, Khaled Diab, **Shivang Aggarwal**, Lianjie Cao, Faraz Ahmed, Puneet Sharma, Sonia Fahmy
ACM EMS 2023
- **Is WiFi 802.11mc Fine Time Measurement Ready for Prime-Time Localization? (Best Paper Award)**
Shivang Aggarwal, Ramanujan K Sheshadri, Karthikeyan Sundaresan, Dimitrios Koutsonikolas
ACM WiNTECH 2022
- **A Detailed Look at MIMO Performance in 60 GHz WLANs**
Shivang Aggarwal, Srisai Karthik Neelamraju, Ajit Bhat, Dimitrios Koutsonikolas
ACM SIGMETRICS / IFIP Performance 2022
- **MuSher: An Agile Multipath-TCP Scheduler for Dual-Band 802.11ad/ac Wireless LANs**
Shivang Aggarwal, Swetank Kumar Saha, Imran Khan, Rohan Pathak, Dimitrios Koutsonikolas, Joerg Widmer
IEEE/ACM ToN 2022
- **Throughput Prediction on 60 GHz Mobile Devices for High-Bandwidth, Latency-Sensitive Applications**
Shivang Aggarwal, Zhaoning Kong, Moinak Ghoshal, Y. Charlie Hu, Dimitrios Koutsonikolas
PAM 2021
- **An Experimental Study of the Performance of IEEE 802.11ad in Smartphones**
Shivang Aggarwal, Moinak Ghoshal, Piyali Banerjee, Dimitrios Koutsonikolas
Elsevier COMCOM 2021
- **802.11ad in Smartphones: Energy Efficiency, Spatial Reuse, and Impact on Applications**
Shivang Aggarwal, Moinak Ghoshal, Piyali Banerjee, Dimitrios Koutsonikolas, Joerg Widmer
IEEE INFOCOM 2021
- **LiBRA: Learning-Based Link Adaptation Leveraging PHY Layer Information in 60 GHz WLANs**
Shivang Aggarwal, Urjit Satish Sardesai, Viral Sinha, Deen Dayal Mohan, Moinak Ghoshal, Dimitrios Koutsonikolas
ACM CoNEXT 2020
- **How to Evaluate Mobile 360° Video Streaming Systems?**
Shivang Aggarwal, Sibendu Paul, Pranab Dash, Nuka Saranya Illa, Y. Charlie Hu, Dimitrios Koutsonikolas, Zhisheng Yan
ACM HotMobile 2020
- **Performance and Pitfalls of 60 GHz WLANs Based on Consumer-Grade Hardware**
Swetank Kumar Saha, **Shivang Aggarwal**, Hany Assasa, Adrian Loch, Naveen Muralidhar Prakash, Roshan Shyamsunder Anantharamakrishna, Daniel Steinmetzer, Dimitrios Koutsonikolas, Joerg Widmer, Matthias Hollick
IEEE TMC 2020
- **A First Look at 802.11ad Performance on a Smartphone (Best Paper Award)**
Shivang Aggarwal, Arvind Thirumurugan, Dimitrios Koutsonikolas
ACM mmNets 2019
- **MuSher: An Agile Multipath-TCP Scheduler for Dual-Band 802.11ad/ac Wireless LANs**
Swetank Kumar Saha, **Shivang Aggarwal**, Dimitrios Koutsonikolas, Joerg Widmer
ACM MobiCom 2019

PATENTS

- **Realizing Enterprise-Grade Localization using WiFi 802.11mc Fine Time Measurement**
US Patent 12,543,019
- **Timestamping Of Direct Wireless Path Signal**
Pending US Patent Application 18/734,867

- **Classification of an AP Pair as a Line-Of-Sight Pair or Non-Line-Of-Sight Pair**
Pending US Patent Application 18/734,839
- **QOE-aware Dynamic Resource Allocation**
Pending US Patent Application 18/903,749
- **Dynamic TDD Policy Adaptation**
Pending US Patent Application 19/035,235
- **Wideband Reconfigurable Metamaterial Surface for Spectrum Sensing and Beamforming**
Pending US Patent Application 19/202,487
- **Wireless Communication Technology Convergence through Continuous Client Device Positioning**
Pending US Patent Application 19/217,470
- **Indoor Positioning and Synchronization using Collaborative Receivers**
Pending US Patent Application 19/361,392

AWARDS

- **Best Paper Award** – ACM WiNTECH 2022, ACM mmNets 2019
- **Best Paper Runner-up Award** – ACM CoNEXT 2024
- **Outstanding PhD Research Award** – Northeastern University College of Engineering 2022
- **Best Dataset Award** – Passive and Active Measurement (PAM) Conference 2021
- **Best PhD Research Award** – UB CSE 2020
- **Best PhD Teaching Award** – UB CSE 2020
- **Best Poster Award** – UB CSE PhD Poster Competition 2019
- **HPE Culture Award** – HPE Office of the CTO Culture Award 2025

SERVICE

- **TPC Co-Chair** – ACM S³ Wireless of the Students, by the Students, and for the Students Workshop (ACM MobiCom 2022)
- **TPC Member** – ACM WiNTECH (2022 - 2024), IEEE WoWMoM (2023 - 2026), IEEE LANMAN (2023), IEEE INFOCOM (2024), IEEE MASS (2024 - 2025), IEEE ICCCN (2024 - 2025), IEEE ICNP (2026)
- **Reviewer** – IEEE/ACM Transactions on Networking (TON), IEEE Transactions on Mobile Computing (TMC), IEEE Transactions on Wireless Communications (TWC), IEEE Transactions on Multimedia (TMM), IEEE Journal on Selected Areas in Communications (JSAC), Elsevier Computer Communications (COMCOM), IEEE Network Magazine, ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)
- **Artifact Evaluation Committee Member** – ACM CoNEXT (2023), ACM MobiCom (2023, 2024), ACM SIGCOMM (2023)
- **Web Chair** – ACM S³ Wireless of the Students, by the Students, and for the Students Workshop (ACM MobiCom 2018)
- **Secretary** – UB Computational Sciences Club 2018-19

SOURCE CODE/DATA RELEASE

- **MuSher** – Released [source code](#) for our MPTCP scheduler, a set of MPTCP instrumentation [tools](#), and the [dataset](#) characterizing MPTCP performance
ACM MobiCom 2019, IEEE/ACM ToN 2022
- **LiBRA, 60 GHz Link Adaptation** – Released [dataset](#) used in studying link adaptation under various link impairment scenarios and developing LiBRA, a learning based 60 GHz link adaptation protocol
ACM CoNEXT 2020, ACM MSWiM 2020
- **60 GHz Throughput Prediction** – Released [dataset](#) used to study throughput predictability of 60 GHz mobile devices for high-bandwidth, latency-sensitive applications
PAM 2021
- **802.11ad Smartphone** – Released [dataset](#) used to study 802.11ad power consumption, spatial reuse, and application performance on smartphones
IEEE INFOCOM 2021
- **MPTCP for mmWave Smartphones** – Released [source code](#) for our MPTCP kernel for a dual-band smartphone (ASUS RoG Phone II) equipped with 802.11ad and 802.11ac radios
ACM WiNTECH 2021
- **MIMO Performance in 60 GHz** – Released [dataset](#) used to study SU- and MU-MIMO performance in 60 GHz WLANs in various environments
ACM SIGMETRICS/IFIP Performance 2022